

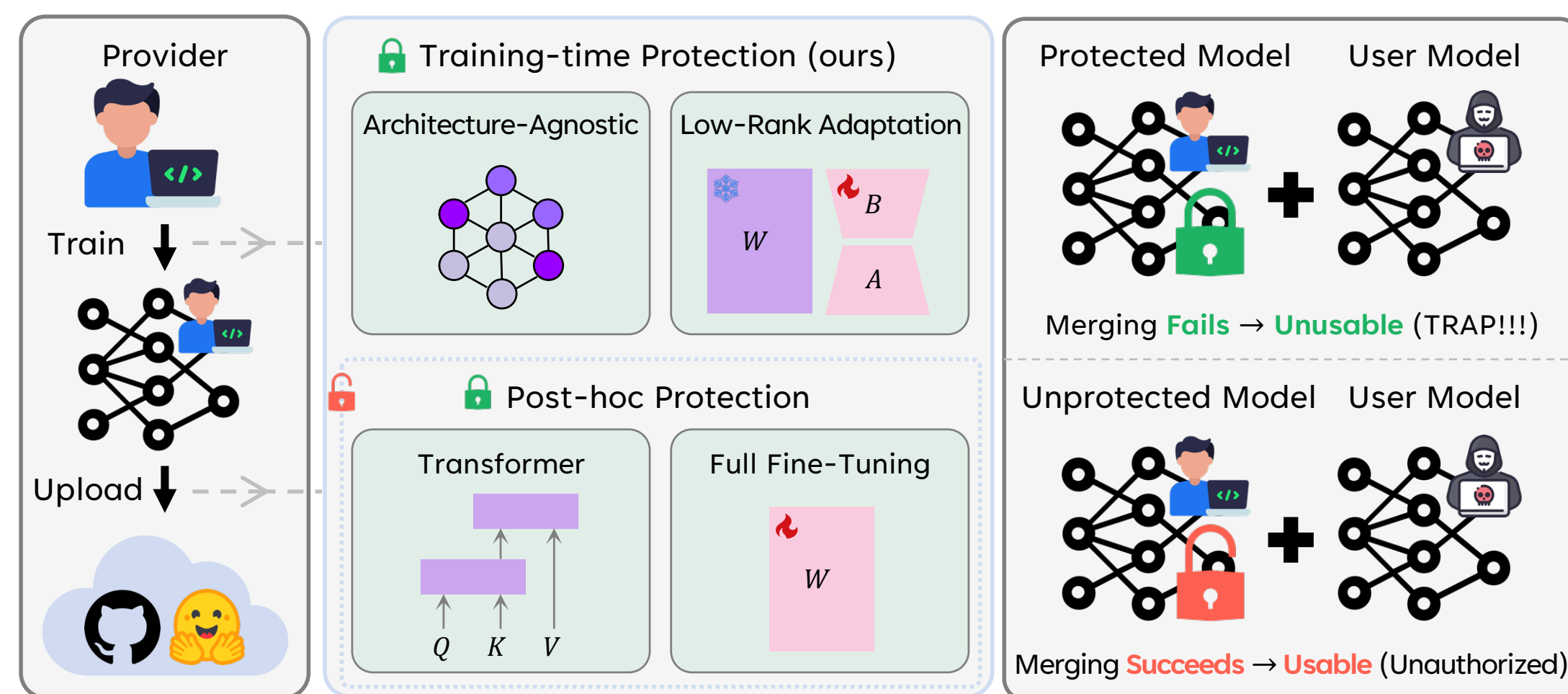
Making Models **Unmergeable** via **Scaling-Sensitive Loss Landscape**

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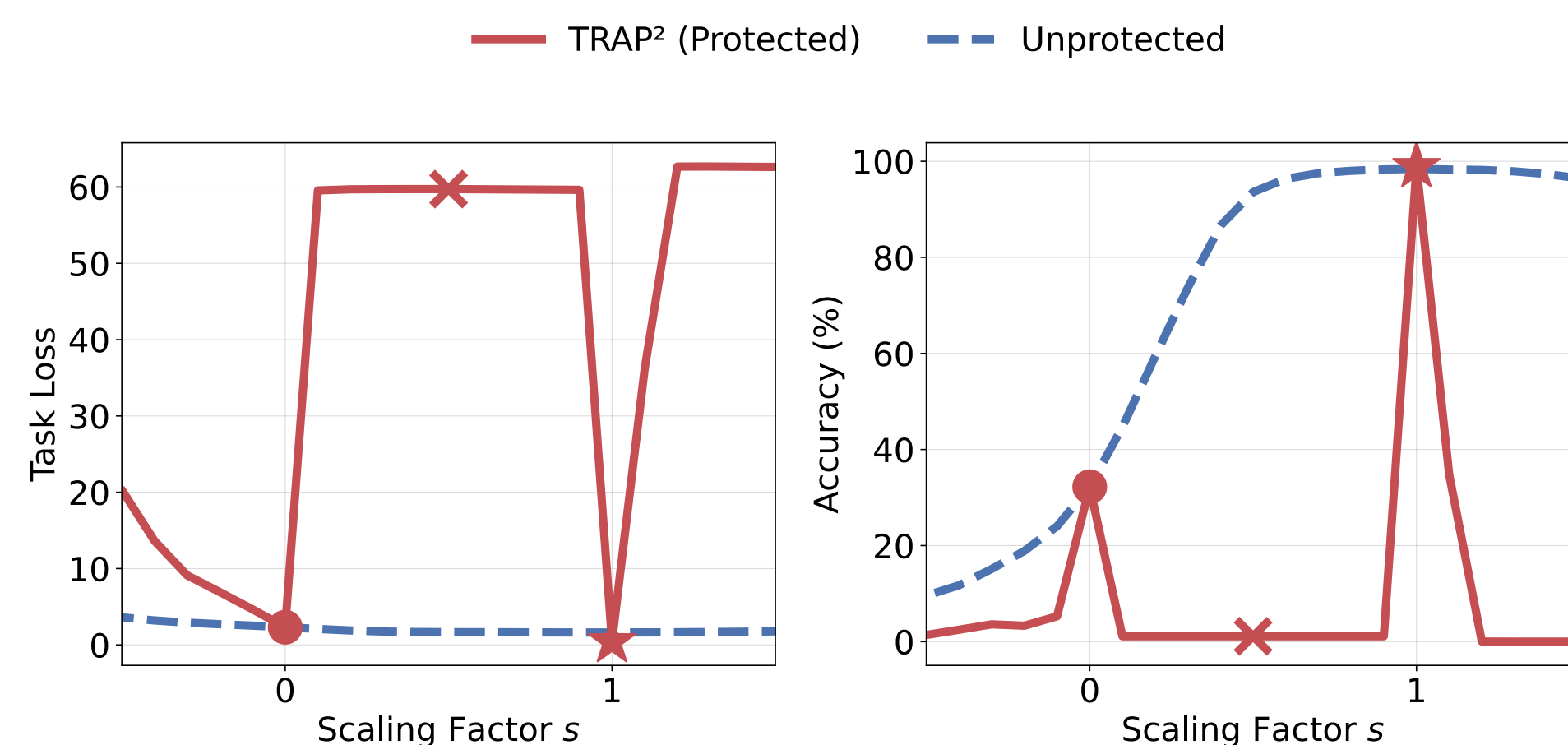
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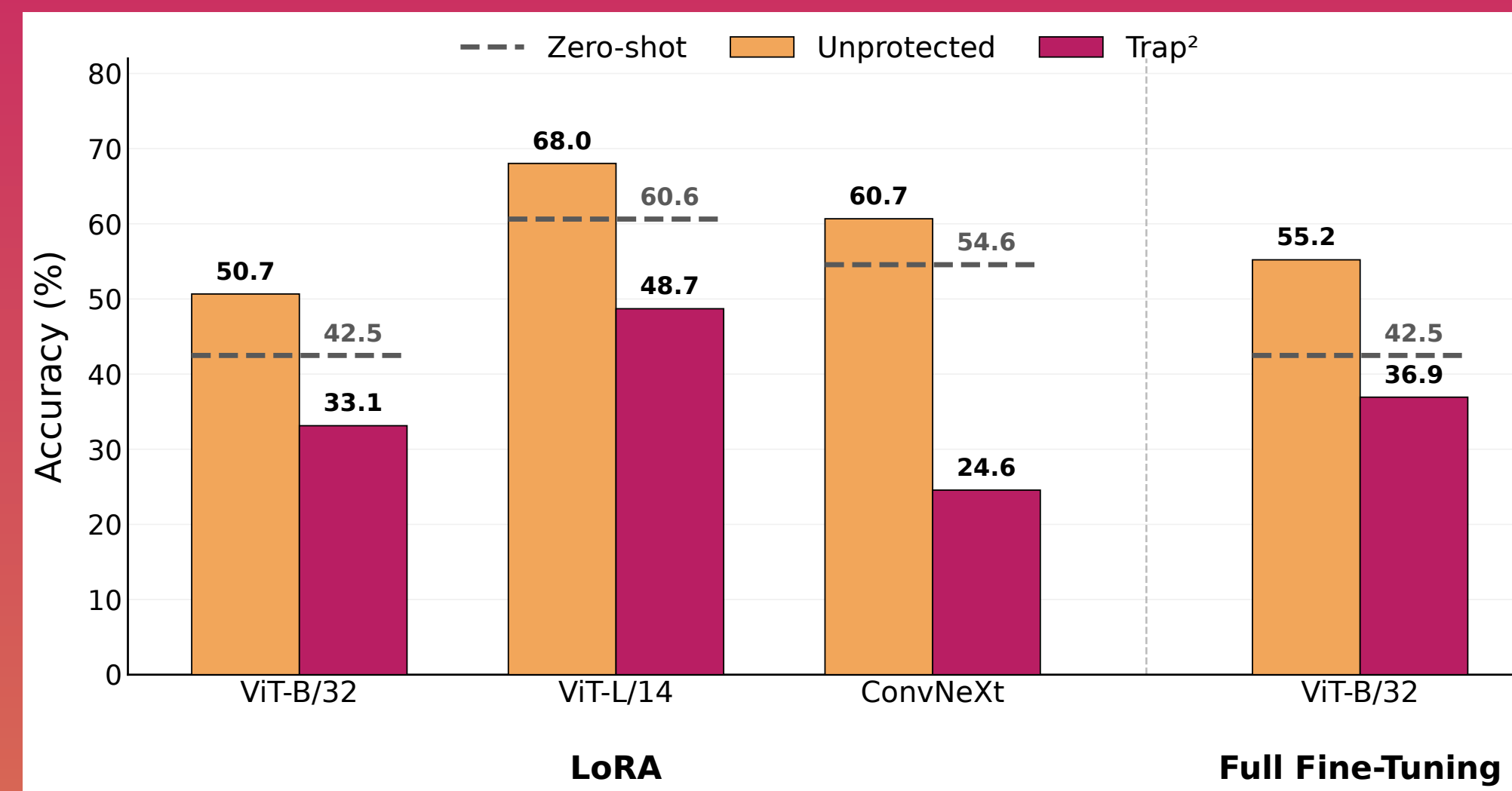
Trap² is the first training-time protection method against **unauthorized merging**, applicable **beyond Transformer** and **beyond full model weights**.



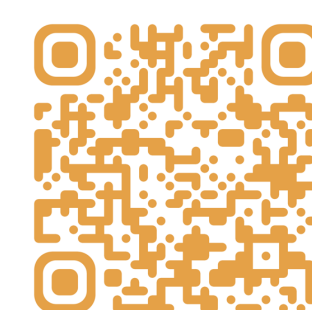
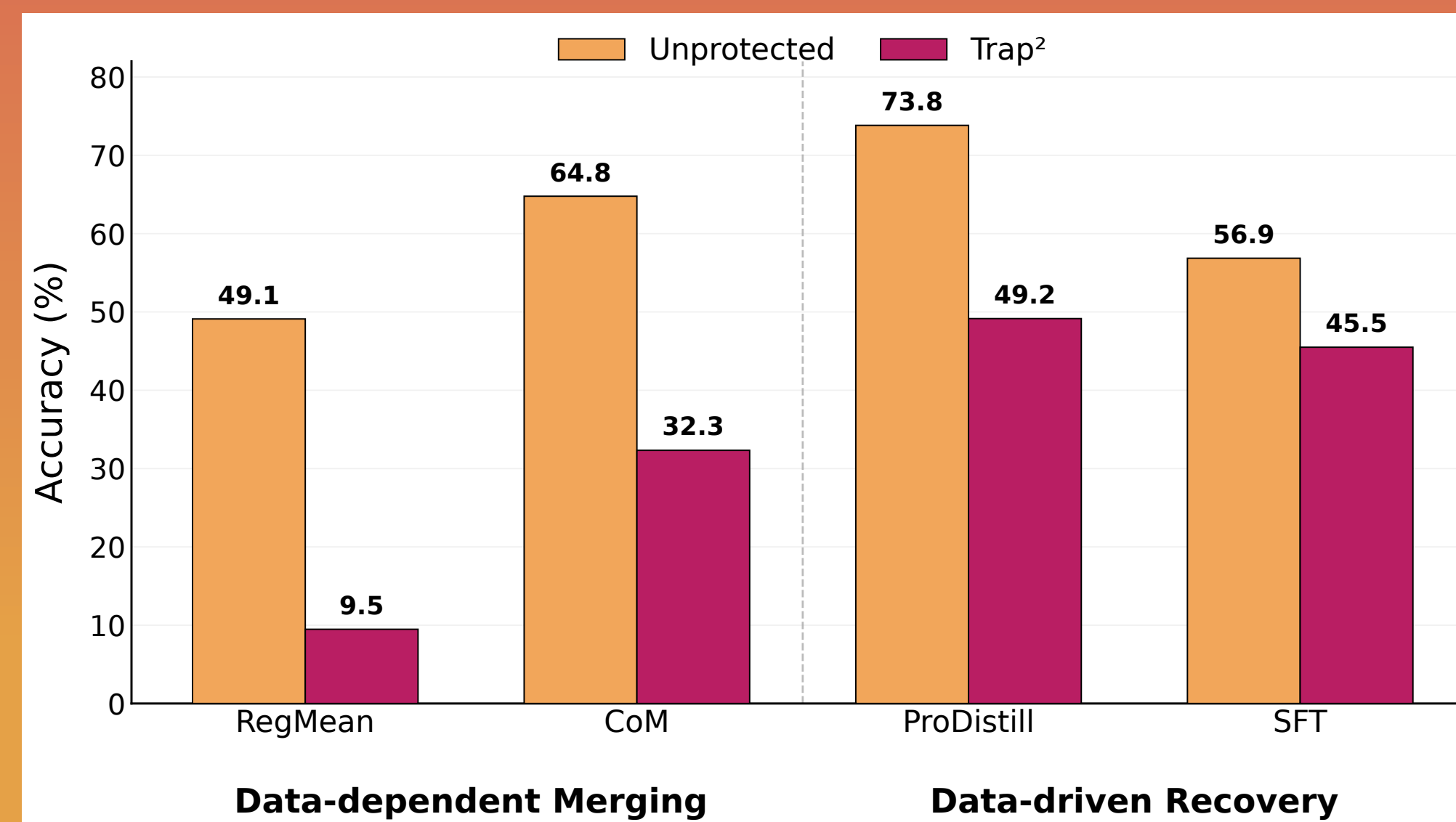
Trap² uses **scaling as a proxy** for merging: stable as intended, **fragile under re-scaling**.



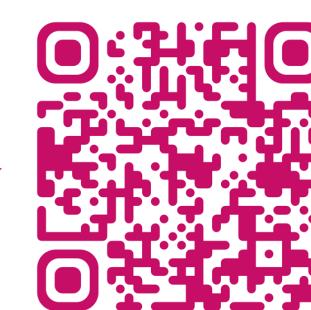
Useful standalone, unreliable when merged.



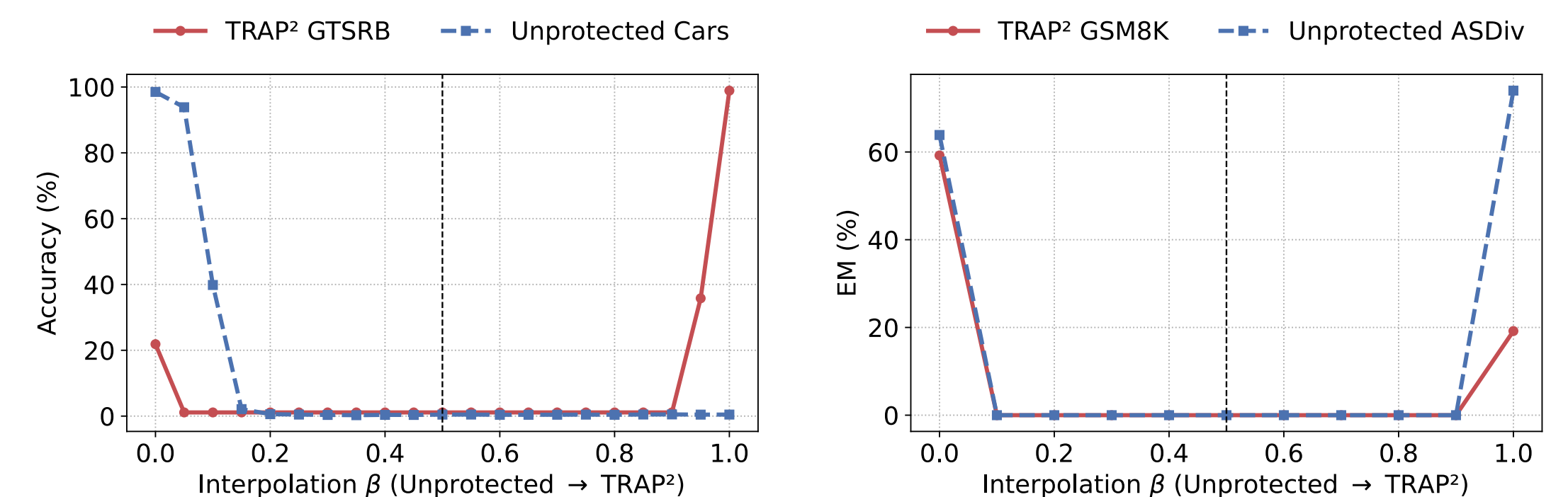
More data, limited escape.



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Two Types of Degradation after merging:
① **Down-scaling** (Trap²) ② **Cross-adapter** (Unprotected)



Trap² objective with LoRA $\Delta W = BA$:

$$J(\Delta W) = \mathcal{L}_{nominal}(\Delta W) - \lambda \cdot \mathcal{L}_{off}(\Delta W)$$

$$\mathcal{L}_{nominal}(\Delta W) := \mathcal{L}(W_0 + \Delta W)$$

$$\mathcal{L}_{off}(\Delta W) := \mathbb{E}_{s \in [s_{min}, 1-\delta] \cup [1+\delta, s_{max}]} \left[\frac{1}{s} \cdot \mathcal{L}(W_0 + s \cdot \Delta W) \right]$$

Intuition: $\mathcal{L}_{nominal}$ preserves performance, while $\mathcal{L}_{off}$ induces sensitivity to re-scalings.

Extension of **Trap²** to Full Fine-Tuning:

Calculate $\mathcal{L}_{nominal}$ at $\Delta W_t := W_t - W_0$ at each step t .

Key Takeaways:

Trap² makes an update **scaling-sensitive**, so that it stays **useful at the nominal scale** but **collapses under off-nominal re-scaling** that merging induces.